

Bugs found: 5

1. Use of freed memory: 'q' is assigned 'p' before 'free(p)', then 'q[2]' is accessed after 'p' is freed, resulting in undefined behavior.
2. Out-of-bounds write: 'q[2]' accesses index 2 of a 2-element array (valid indices 0 and 1), writing beyond the allocated bounds.
3. Memory leak: 'x' is allocated with 'malloc(10)' but never freed before the function returns.
4. Dangling pointer: After 'free(p)', both 'p' and 'q' point to freed memory; neither is set to NULL, leaving dangling pointers.
5. Incorrect allocation size: 'malloc(10)' allocates 10 bytes rather than a specific number of elements (e.g., 'malloc(10 * sizeof(int))'), making the intent ambiguous and potentially incorrect.